PharmaAI Research Report

Montelukast - Innovation Opportunity Assessment

Report Generated: December 17, 2025 at 18:00

Executive Summary

Decision: GO
Confidence Score: 85/100
Recommendation: ✓ Market shows attractive size and growth | ✓ Patent landscape favorable for FTO | ✓ Clinical space has room for innovation

1. Market Intelligence (IQVIA)

Market Size: $850M USD
Growth Rate (CAGR): 5.2%
Therapy Area: Respiratory
Competition Intensity: Medium (Mixed brand/generic)
Attractiveness Score: 28.8/100

Top Competitors

• Singulair (Merck): 35% market share
• Generic Montelukast: 45% market share
• Others: 20% market share

2. Patent Landscape

Active Patents: 2
FTO Risk Score: 6.5/10
Risk Level: Medium
Recommendation: Conduct detailed FTO study; design-around may be needed

Key Patents

• **US8234567**: Pharmaceutical compositions comprising montelukast (Merck & Co.)
• **US7654321**: Methods of treating asthma with leukotriene inhibitors (Generic Pharma Inc.)

3. Clinical Trials Landscape

Total Active Trials: 8
Saturation Level: Low
Opportunity Score: 9/10

Indication Diversity

• Asthma
• COVID-19
• Chronic Urticaria
• Allergic Rhinitis

Repurposing Opportunities

• **Asthma**: Active clinical trials exploring Asthma
• **COVID-19**: Active clinical trials exploring COVID-19
• **Chronic Urticaria**: Active clinical trials exploring Chronic Urticaria

4. Trade Intelligence (EXIM)

Sourcing Risk: Medium
Top Supplier: India (47.4%)

Top Exporters

• India: 450 tonnes
• China: 380 tonnes
• Israel: 120 tonnes

5. Internal Knowledge Insights

Documents Found: 2

• Montelukast shows declining market share in traditional asthma segment
• Internal feasibility study on Montelukast extended-release formulation

6. Web Intelligence

Recent Scientific Publications

• **Montelukast, a once-daily leukotriene receptor ... - PubMed** (Unknown, 2024)
• **Montelukast, a Once-Daily Leukotriene Receptor Antagonist, in ...** (Unknown, 2024)

Clinical Guidelines

• Montelukast Monograph for Professionals - Drugs.com - Unknown
• Evaluation of the effect of montelukast drug in improving the - Unknown

Final Recommendations

IQVIA Analysis:
{'molecule': 'Montelukast', 'therapy_area': 'Respiratory', 'market_size_usd_million': 850, 'cagr_percent': 5.2, 'competition_intensity': 'Medium (Mixed brand/generic)', 'top_competitors': [{'name': 'Singulair (Merck)', 'market_share': 35}, {'name': 'Generic Montelukast', 'market_share': 45}, {'name': 'Others', 'market_share': 20}], 'regional_breakdown': {'North America': 450, 'Europe': 250, 'Asia Pacific': 100,

'Rest of World': 50}, 'market_attractiveness_score': 28.8, 'data_source': 'IQVIA (via ZenRows)', 'summary': 'Market Analysis for Montelukast: $850M market growing at 5.2% CAGR. Medium (Mixed brand/generic) competition intensity. Primary therapy area: Respiratory.'}

PATENT Analysis:
{'molecule': 'Montelukast', 'active_patents_count': 2, 'active_patents': [{'patent_number': 'US8234567', 'title': 'Pharmaceutical compositions comprising montelukast', 'assignee': 'Merck & Co.', 'filing_date': '2015-03-12', 'expiry_date': '2035-03-12', 'status': 'Active', 'claims': 15}, {'patent_number': 'US7654321', 'title': 'Methods of treating asthma with leukotriene inhibitors', 'assignee': 'Generic Pharma Inc.', 'filing_date': '2012-08-20', 'expiry_date': '2032-08-20', 'status': 'Active', 'claims': 8}], 'data_source': 'USPTO (patent_client)', 'expired_patents_count': 12, 'fto_risk_score': 6.5, 'fto_assessment': {'risk_level': 'Medium', 'fto_score': 6.5, 'recommendation': 'Conduct detailed FTO study; design-around may be needed', 'suggested_action': 'Engage patent attorney for detailed analysis', 'blocking_patents': []}, 'expiring_within_3_years': [], 'recent_filing_activity': 3, 'competitive_landscape': {'dominant_patent_holders': [{'company': 'Merck & Co.', 'patent_count': 1}, {'company': 'Generic Pharma Inc.', 'patent_count': 1}], 'filing_activity_trend': 'Stable'}, 'summary': 'Patent Landscape for Montelukast: 2 active patents identified. FTO risk score: 6.5/10 (Medium risk). Conduct detailed FTO study; design-around may be needed'}

CLINICAL_TRIALS Analysis:
{'data_source': 'ClinicalTrials.gov (pytrials)', 'molecule': 'Montelukast', 'total_active_trials': 8, 'trials_sample': [{'nct_id': 'NCT04567890', 'title': 'Montelukast in COVID-19 Related Respiratory Distress', 'sponsor': 'Johns Hopkins University', 'phase': 'Phase 2', 'status': 'Recruiting', 'enrollment': 120, 'start_date': '2023-05-15', 'indication': 'COVID-19'}, {'nct_id': 'NCT03891234', 'title': 'Efficacy of Montelukast in Chronic Urticaria', 'sponsor': 'Mayo Clinic', 'phase': 'Phase 3', 'status': 'Active', 'enrollment': 250, 'start_date': '2022-11-20', 'indication': 'Chronic Urticaria'}, {'nct_id': 'NCT05123456', 'title': 'Montelukast for Exercise-Induced Asthma', 'sponsor': 'University of California', 'phase': 'Phase 2', 'status': 'Recruiting', 'enrollment': 85, 'start_date': '2024-01-10', 'indication': 'Asthma'}], 'phase_distribution': {'Phase 1': 1, 'Phase 2': 4, 'Phase 3': 2, 'Phase 4': 1}, 'indication_diversity': ['Asthma', 'COVID-19', 'Chronic Urticaria', 'Allergic Rhinitis'], 'saturation_analysis': {'saturation_level': 'Low', 'interpretation': 'Low clinical activity; potential white space opportunity', 'opportunity_score': 9, 'total_active_trials': 8}, 'repurposing_opportunities': [{'new_indication': 'Asthma', 'rationale': 'Active clinical trials exploring Asthma', 'confidence': 'Low'}, {'new_indication': 'COVID-19', 'rationale': 'Active clinical trials exploring COVID-19', 'confidence': 'Low'}, {'new_indication': 'Chronic Urticaria', 'rationale': 'Active clinical trials exploring Chronic Urticaria', 'confidence': 'Low'}], 'top_sponsors': [{'sponsor': 'Johns Hopkins University', 'trial_count': 1}, {'sponsor': 'Mayo Clinic', 'trial_count': 1}, {'sponsor': 'University of California', 'trial_count': 1}], 'summary': 'Clinical Trials Analysis for Montelukast: 8 active trials identified. Saturation level: Low. Trials span 4 different indications, suggesting repurposing potential.'}

EXIM Analysis:
{'molecule': 'Montelukast', 'top_exporters': [{'country': 'India', 'volume_tonnes': 450, 'value_usd_million': 85}, {'country': 'China', 'volume_tonnes': 380, 'value_usd_million': 72}, {'country': 'Israel', 'volume_tonnes': 120, 'value_usd_million': 28}], 'top_importers': [{'country': 'USA', 'volume_tonnes': 520, 'value_usd_million': 110}, {'country': 'Germany', 'volume_tonnes': 180, 'value_usd_million': 38}, {'country': 'Brazil', 'volume_tonnes': 95, 'value_usd_million': 20}], 'trade_trends': {'year_2023': 185, 'year_2024': 195, 'growth_percent': 5.4}, 'sourcing_risk_assessment': {'risk_level': 'Medium', 'reason': 'Moderate concentration in India', 'top_supplier': 'India', 'top_supplier_share_percent': 47.4}, 'import_dependency_countries': ['USA'], 'data_source': 'UN Comtrade API', 'summary': 'EXIM Analysis for Montelukast: Top exporter is India, top importer is USA. Trade growing at 5.4% annually. Sourcing risk: Medium (Moderate concentration in India).'}

INTERNAL_KNOWLEDGE Analysis:

{'molecule': 'Montelukast', 'documents_found': 2, 'relevant_documents': [{'doc_id': 'INT-2023-045', 'source': 'Strategic Planning Deck Q3 2023', 'content_snippet': 'Montelukast shows declining market share in traditional asthma segment. Opportunity identified in repurposing for chronic urticaria based on emerging clinical data. Preliminary market analysis suggests $200M+ opportunity.', 'relevance_score': 0.92, 'document_type': 'Strategy Report'}, {'doc_id': 'INT-2022-128', 'source': 'R&D; Pipeline Review 2022', 'content_snippet': 'Internal feasibility study on Montelukast extended-release formulation. Technical team confident in developing once-daily formulation. Patent filing recommended by Q1 2024.', 'relevance_score': 0.85, 'document_type': 'R&D; Report'}], 'key_insights': ['Montelukast shows declining market share in traditional asthma segment', 'Internal feasibility study on Montelukast extended-release formulation'], 'summary': 'Found 2 relevant internal documents for Montelukast (avg relevance: 0.89). Key sources include internal strategy reports, R&D; pipeline reviews, and feasibility studies.'}

WEB_INTELLIGENCE Analysis:
{'molecule': 'Montelukast', 'disease': None, 'search_queries': ['Montelukast clinical guidelines', 'Montelukast latest research publications', 'Montelukast repurposing opportunities'], 'guidelines': [{'title': 'Montelukast Monograph for Professionals - Drugs.com', 'source': 'Unknown', 'url': '', 'excerpt': '... at medium or high dosages and adjunctive therapy with either a long-acting inhaled β 2 -agonist or montelukast is the only preferred treatment .....', 'relevance_score': 0.85}, {'title': 'Evaluation of the effect of montelukast drug in improving the', 'source': 'Unknown', 'url': '', 'excerpt': 'All patients received treatments according to the national guidelines , while the case group additionally applied 10 mg montelukast for 14 days....', 'relevance_score': 0.85}, {'title': ''Montelukast will carry 'more prominent warnings' of'', 'source': 'Unknown', 'url': '', 'excerpt': 'Further safety measures will be introduced for montelukast , a drug used for the treatment of asthma, to ensure patients are informed of the risk of .....', 'relevance_score': 0.85}], 'scientific_publications': [{'title': 'Montelukast, a once-daily leukotriene receptor ... - PubMed', 'authors': 'Various', 'journal': 'Unknown', 'url': '', 'abstract': 'Jun 8, 1998 · Montelukast , compared with placebo, significantly improved asthma control during a 12-week treatment period. Montelukast was generally well tolerated, with an adverse event profile comparable with that of placebo....', 'publication_date': '2024', 'citations': 98, 'relevance_score': 0.88}, {'title': 'Montelukast, a Once-Daily Leukotriene Receptor Antagonist, in ...', 'authors': 'Various', 'journal': 'Unknown', 'url': '', 'abstract': 'This clinical trial demonstrates that montelukast provided clinical benefit during the 12-week treatment period by consistent and significant improvement of all asthma control variables compared with placebo....', 'publication_date': '2024', 'citations': 133, 'relevance_score': 0.88}, {'title': 'Montelukast Use and the Risk of Neuropsychiatric Adverse ...', 'authors': 'Various', 'journal': 'Unknown', 'url': '', 'abstract': 'Apr 1, 2025 · Objective: To assess the potential association between the use of montelukast and the risk of neuropsychiatric adverse events in children and adolescents. Design, setting, and participants: This nationwide register-based cohort study used data from Sweden from January 1, 2007, to Novem...', 'publication_date': '2024', 'citations': 193, 'relevance_score': 0.88}], 'recent_news': [{'title': 'FDA Approves New Indication for Montelukast', 'source': 'PharmaTimes', 'url': 'https://pharmatimes.com/montelukast-approval', 'snippet': 'The FDA has granted approval for Montelukast to be used in treatment of new indication, expanding its therapeutic applications.', 'published_date': '2024-02-15', 'relevance_score': 0.9}, {'title': 'Generic Competition Heats Up for Montelukast', 'source': 'FiercePharma', 'url': 'https://fiercepharma.com/montelukast-generics', 'snippet': 'Three new generic manufacturers enter the Montelukast market, putting pressure on branded products.', 'published_date': '2024-01-20', 'relevance_score': 0.75}], 'patient_forums': [{'source': 'PatientsLikeMe Forum', 'url': 'https://patientslikeme.com/conditions/montelukast', 'insight': 'Patients report positive outcomes with Montelukast for various conditions. Common theme: improved quality of life with minimal side effects.', 'sentiment': 'Positive', 'sample_size': '~500 patient reports', 'relevance_score': 0.7}], 'summary': 'Web Intelligence for Montelukast: Found 3 clinical guidelines, 3 scientific publications, and 2 recent news articles. Evidence supports both traditional and emerging indications.'}

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